

Data Lineage

In this train, we'll explore the significance of monitoring data evolution over time and delve into the concept of data lineage so that we can ensure data integrity and enhance the efficiency of data and machine learning projects.

Learning Objectives

By the end of this train, you should be able to:

- Understand what data lineage is.
- Follow best practices when implementing data lineage.
- Familiarise yourself with available tools for data lineage.

What is Data Lineage?

Introduction to Data Lineage

In an age where data are abundant, **data integrity** is of utmost importance. Data integrity ensures that the quality and consistency of our data are up to standard for the entire lifecycle of the data. The **data lifecycle** refers to the movement of data from a source system to a sink location — and we need to monitor this to establish **traceability**.

Data lineage will help us track where the data came from, what changes have occurred, and where the data are going. Think of data lineage as visualising our **data flow** and providing answers to the who, what, where, when, why, and how questions related to the data.

MetaData

Data lineage is one of many aspects covered under data management, including:

- Data governance
- Data dictionary
- Data catalog
- Data provenance

Levels of Data Lineage

Data lineage can be split into two levels: **business data lineage** and **technical data lineage**.

Business users need to trace errors back to the data source and ensure accuracy. Business data lineage provides information on the origins, movement, and transformations

of data over time — focusing on **what**, **where**, and **when** the data changed.

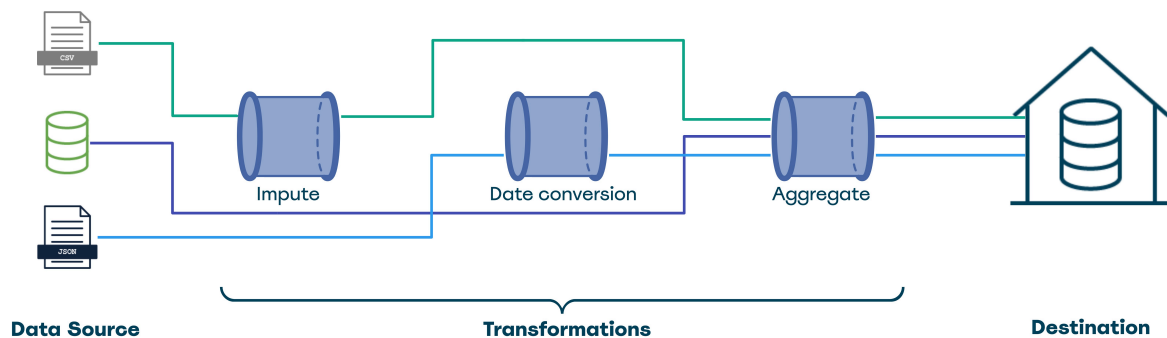
Technical users need to see the inner workings of code or pipelines and how it influences data flow. They document all elements related to compliance, procedures, and transformations — focusing on **how** data has changed, **who** can access it, and **how** it is shared.

Example: ETL Pipeline Data Lineage



Let's look at a basic data lineage example of the ETL process. Data is ingested from different data sources (CSV, database, JSON), undergoes transformations, then written to a data warehouse.

Data Flow Diagram



ETL data flow diagram

The transformations are:

- Transformation 1: Imputation of missing values (CSV source)
- Transformation 2: Date conversion (JSON source)
- Transformation 3: Aggregation (all three sources)

To track the origin, transformations, and when/where they occurred, we document these attributes:

- Input data sources
- Physical location of the data
- Owner of the data
- Created time
- Number of records

```
[{
  "input_source": "csv",
  "location": "/hdfs/data/wfm.db/csv_ingestion",
  "owner": "John Doe",
  "created_time": "2021-01-01T13:01:02Z",
  "number_records": "21902"
},
{
  "input_source": "json",
  "location": "/hdfs/data/wfm.db/json_ingestion",
  "owner": "Jill Deer",
  "created_time": "2021-01-01T14:04:02Z",
  "number_records": "1902"
},
{
  "input_source": ["postgres_ingestion", "csv_imputation", "json_date_conversion"],
  "location": "/hdfs/data/wfm.db/aggregation_table",
```

```
"owner": "John Deer",  
"created_time": "2021-01-01T15:20:00Z",  
"number_records": "73678"  
}]
```

Good to know: We have metadata at each node in the processing pipeline. While creating a map of data lineage, we will be able to recreate the data processing pipeline in the process as well.

Best Practices for Data Lineage

1. Have technical audit trails



Technical audit trails verify and track sensitive data, logging: where the data originated from, how it was changed, the quality of the data, who can access it, and how it was shared.

Audit trails are necessary to ensure data comply with data protection laws (GDPR, CCPA, HIPAA, COPPA, POPI).

2. Test your data quality



Having poor data can have severe effects on businesses. Data quality is improved by having lineage in place that allows organisations to track data errors. Data quality rules can be instituted to allow users to calculate data quality scores, improving the trustworthiness of the data.

3. Have technical and business visibility



Data lineage needs to cater to the needs of all users. Having complete visibility allows business users to understand the impact of a decision before making it. For technical users, data lineage aids in finding the cause of data quality issues and allows effective response.

4. Have a data catalog



A data catalog is an inventory of all data assets within an organisation, organised to make referencing easy. Data lineage and data quality within a data catalog provide transparency, making the data flow clearly visible and ensuring trustworthiness.

5. Automate data lineage



Use an automated tool for your data lineage. Automating this process ensures that your data lineage is consistent and always up to date.

Data Lineage Tools

The following are all-in-one solutions for data governance and metadata management with a data lineage component. These seven tools are currently reported to be the best available:

Azure Purview

A unified data governance service that helps manage on-premises, multi-cloud, and SaaS data. Provides automated data discovery, sensitive data classification, and end-to-end data lineage. Three main functions: Data map (tracks data and shows lineage), Data catalog (shows what data exists and where), Data insights (reports to understand the data).

Apache Atlas

Provides open metadata management and governance capabilities for organisations to build a catalog of their data assets, classify and govern these assets, and enable collaboration around data for scientists, analysts, and governance teams.

Collibra

An enterprise-oriented data governance platform for data management and stewardship. Collibra Lineage integrates with Collibra Catalog and automatically maps the movement of data, providing complete visibility.

Keboola

Automates the collection, transformation, and storage of data. All metadata are tracked and logged automatically.

Dremio

Query data stored in AWS and Azure data lakes. Constructs a data flow diagram to track data sources, datasets, and queries. Available for free trial.

Kylo

An open-source data lake management software platform with features including: data ingestion, data wrangling, data and metadata discovery, data lake monitoring, and pipeline design.

Octopai

Purely focused on data lineage with no additional functionality. Connects to your data storage and extracts relevant metadata required for data lineage insights.

Activity: Take some time to research the steps and processes required to build your own data lineage tool.

Conclusion

Data lineage plays a critical role in ensuring **data integrity** and **transparency** throughout the data lifecycle. By implementing **best practices** and leveraging **automated tools** like Azure Purview or Apache Atlas, businesses can effectively track and manage data flow, ensuring its quality and trustworthiness while complying with regulatory standards.

Key Takeaways



- **Takeaway 1:** Data lineage tracks the flow, transformations, and origins of data throughout its lifecycle, ensuring integrity, transparency, and traceability.
- **Takeaway 2:** Best practices include maintaining audit trails, testing data quality, ensuring both technical and business visibility, using a data catalog, and automating lineage tracking.
- **Takeaway 3:** Automation is key for keeping lineage up to date, reducing manual work, and adapting to changes in data pipelines.
- **Takeaway 4:** Tools like Azure Purview, Apache Atlas, and Collibra provide end-to-end solutions for governance, metadata management, and lineage visualisation, aiding compliance and decision-making.